जय महादेव™

HC HARIHAR CHEMICALS

Since 1997

ISO 9001:2015

HARIHAR CHEMICALS

NAPHTHOLS, BASES & DYES





Manufacturer & Exporter

NAPHTOLS, BASES & DYES

HARITHOLS AND FAST BASES

The **Harihar Chemicals** together with Fast Bases, form a series of products which are used for production of insoluble azo-colours on cellulosic material such as cotton, linen, viscose, rayon, jute, hemp etc. They yield a wide range of yellow, golden yellow, orange, scarlet, red, maroon etc having good to excellent fastness properties Any Harithol can be combined with any Fast Base to get a variety of azoic pigments on the fibre selection of the particular combination to be used in any instance is determined, in practice, by the shade and fastness property of the resulting dyeing and secondly by the suitability of the combination for use under the proposed condition of application.

DYEING PROCESS

The dyeing process consists of the following four stages.

1. Impregnation of the material in an Harithol solution.
2. Removal of the excess Harithol solution.
3. Development of the impregnated material in a solution of diazotised Fast Base.
4. After treatments.

Pretreatment : The material for impregnation whether in yam form or piece goods should be given a scouring treatment at boil with Soda ash. Caustic soda and a detergent such as Harithol T pdr. H/Conc. followed by thorough washing and drying.

1. Preparation of Harithol solution & Impregnation

Two methods are available for dissolving Harithols.

a) Hot dissolving method:

In this method Harithols are dissolved by pasting with equal quantity of Turkey Red Oil and water to get a smooth consistency and then dissolved by the addition of required quantity of caustic soda and boiling water. The resulting contents are boiled, if necessary, to get a clear solution. If Formaldehyde is to be used, then it is added to this concentrated solution after cooling to room temp and made upto the required volume of the impregnation bath with water containing TRO and caustic soda Impregnation is carried out generally for 20-30 minutes.

The depth of the shade obtained depends on the amount of Harithol taken up by the fibre and this is governed by the following factors

1. Concentration of the Harithol in bath.
2. Substantivlty of the Harithol.
3. Time of impregnation.
4. Temperature of the bath (Harithol Solution)
5. Material to Liquor ratio.
6. Addition of salts.

Quantity of chemicals required to prepare Harithol solution and substantivity of Harithols

For 1 Kg Harithol	TRO 50% Kgs	Hot water Litres (66°Tw) Litres	Caustic Soda (40%) Litres	Formal-dehyde salt that be added gm/l	Maximum Common	Substantivity Approximate
AS	1	15	1.6	1	50	10%
AS-D	1	15	1.6	1	50	10%
AS-OL	1	15	1.6	0.5	50	13%
AS-BS	1	15	1.6	1	30	18%
AS-BO	1	20	2.2	1	30	23%
AS-G	1	15	2.7	NIL	25	23%
AS-TR	1	30	2.2	0.5	20	20%
AS-SW	1	30	4.0	0.5	10	40%

b) Cold dissolving method :

In this method the required quantity of Harithol is placed in a tall narrow continer (to minimise evaporation of spirit) an pasted with methylated spirit overwhich caustic soda is added. Harithol is instantaneously dissolved on addition of cold water while stirring. Formaldehyde if needed can be added at this stage and after 10 minutes the solution so obtained is diluted with water containing the required quantity of Turkey red oil and caustic soda for impregnation.

Quantity of chemicals for cold dissolving of Harithols

For 1 Kg of Harithol	Methylated spirit Litres	Caustic soda (66°Tw) Litres	Cold water Litres	Formaldehyde 40% Litres
AS	1	0.55	1.5	0.5
AS-D	1	0.55	1.5	0.5
AS-OL	1	0.55	1.5	0.5
AS-SW	1.25	0.44	1.0	0.5
AS-BS	1.5	0.55	1.5	0.5
AS-TR	1.5	0.55	2.0	1.5
AS-BO	1.75	0.66	1.3	0.5
AS-G	1	0.77	2.0	NIL

Note : Recommended quantity of Turkey red oil and caustic soda in final impregnation bath :

Turkey Red Oil 50%: TRO is used as a wetting agent while dissolving Harithols. The addition of TRO maintains the stability of the Harithol solution and assists in the wetting of the material and ensures thorough penetration resulting in the even dyeing of the material Therefore a minimum of 4 ml/1 TRO for all type of dyeing, 12.5 ml/l for padding and 20 ml/1 for tub liquoring should be used.

Caustic Soda (66°Tw) : Caustic soda is used for converting the Harithol to sodium salt of Harithol while dissolving, more quantity is required in the final bath, otherwise sodium naphtholate so formed may get hydrolised resulting in precipitation of Harithol. It also protects the impregnated material during exposure to air .The following minimum quantities of caustic soda are required in the final bath

45 ml/1 for pale shades
66 ml/1 for medium shades
8.8 ml/1 for full shades

For Harithol AS-G:

27 ml/1 for pale shades.
6.8 ml/1 for medium to heavy shades.

Formaldehyde 40% : Formaldehyde reacts with dissolved Harithol to form condensation products which are more stable to air and less sensitive to water drops and acid vapours Formaldehyde should not be used in the following cases

1. In the dyeing of Harithol AS-G
2. If the material is dried after impregnation
3. If the dyed material is to because in the discharge, style of printing
4. Formaldehyde should not be used while Harithol solution is hot as it may lead to precipitation

Common Salt: Addition of common salt helps exhaustion However the quantities mentioned should not be exceeded in any case as it may lead to precipitation of Harithol and resulting in weaker dyeing It is preferable to use double the quantity of Glauber's salt or equal quantity of calcined Glauber's salt as common salt may contain salt of magnesium and lime which causes partial precipitation of Harithol, resulting in weaker dyeing.

2) Removal of excess Harithol solution :

After impregnation the material should be thoroughly squeezed or hydroextracted. The fastness to rubbing of the final shade depends to a large extent upto the elliciency of this stage of the process.

3) Development in a solution of selected Fast Base :

The formation of colour takes place during this stage. The impregnated material is dipped in or passed through a bath containing diazotised Fast Base solution.

Diazotisation method 1 : Fast bases which are soluble in water or in dilute hydrochloric acid are best diazotised in the form of their hydrochloride solution. First the required amount of Base is pasted with hydrochloric acid and a little water. Lumps or gritty particles if any should be grinded well at this stage. Hot water is poured over this smooth paste and stirred to complete the dissolution. This solution is cooled to be required temperature and predissolved sodium nitrite (1:1) is added in a thin stream whilst stirring. After keeping the solution for 20 to 30 minutes with occasional stirring sodium acetate dissolved in water is added to neutralise any excess acid. The solution is filtered and made to the required volume with cold water. Finally alkali binding agent Acetic acid 50% or Aluminium sulphate solution is added just before development. Development of the material is carried out for 20-30 minutes.

Quantity of chemicals for diazotisation by method 1.
Table 1

For 1 Kg Fast Base	Hydrochloric Acid Conc. 32° Tw Litres	Hot Water Litre	Sodium Nitrite Kgs.	Temp °C	Sodium Acetate Crystals Kgs.	Acetic Acid 50% Litres	or Aluminium Sulphate Kg.
Yellow GC	1.2	15	0.5	5-10	1	0.75	1
Orange GC	1.2	15	0.5	5-10	1	0.75	1
Scarlet G	2.0	25	0.5	10	1	0.75	1
Scarlet GGS	2.0	10	0.36	10-15	2.5	---	---
Scarlet RC / Scarlet RCR	1.0	20	0.4	15	0.75	0.5	0.75
Scarlet R	2.0	15	0.8	10	1.5	1.0	1.5
Red R Flakes*	2.0	20	1.0	10	1.5	0.5	2.0
Red RC*	1.0	15	0.5	10	0.75	0.25	1
Red TR	1.0	20	0.4	10	0.75	0.15	0.5
Red KB	1.0	25	0.4	10	0.75	0.55	0.75
Garnet GBC	1.6	25 (cold)	0.32	5	1.25	---	---
Blue BB*	1.0	20	0.25	18	0.5	---	---
Blue B*	2.5	20	0.65	12	1.25	---	---

***See note 2**

Diazotisation method 2 :

This method is used for diazotising the Fast Bases mentioned in the Table No. 2. Their hydrochloride are not easily soluble in water. The Fast Base is pasted with a little water and Sodium nitrite solution (1:1) is then added. Hot or cold water is added with stirring. The mixture is added to solution of hydrochloric acid 32°Tw with continuous stirring manintaining the temperature in the range of 10-15°C using ice. After keeping the solution for about 20 min. the excess hydrochloric acid is neutralised with sodium acetate crystal and alkali binding agent such as acetic acid or aluminium sulphate are then added. The solution is then filered and made up to the required volume.

Quantity of chemicals for diazotisation by method 2.
Table 2

For 1 Kg Fast Base	Water Litres	Sodium Nitrite Kgs.	Cold Water Litres	HCl 32° Tw Litre	Sodium Acetate Kgs.	Acetic Acid 50% Litres	or Aluminium Sulphate Kg.
Orange GR	1.5 RT	0.55	30	2.2	1.2	0.7	1
Red B	1.5 (H)	0.50	20	1.8	0.9	0.8	1
Bordeaux GP	1.5 (H)	0.50	20	1.8	0.9	0.8	1
Violet B*	2.5 (H)	0.3	15	11.5	0.55	---	---

***See note 2**

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NOTES

1. Addition of Common salt :

About 20 gm/l Common salt can be added in all developing baths to Imp the rubbing fastness. This addition can be omitted for development of dried ; and in highly concentrated developing bath. In the case of Fast Garnet GEC base only 10 gm/l of salt is required.

2. In the final bath of Fast Red TR, Blue B, Blue BB & Violet B bases 10 gm/l of sodium acetate should be added before development. Similarly an additional quantity of 5-7 gm/l of sodium acetate is to be added in the development bath with Fast Red R Flakes and Fast Red RC base.
3. In all AS-G combinations, before developing the bath is replenished with an addition of 4cc/l Acetic Acid 50% to avoid dull and uneven shades.

Aftertreatments :

After development the material is rinsed in cold water, followed by souring with 3cc/l hydrochloric acid at 40°C for 10 minutes. The material is rinsed well and soaped in a bath containing

1-2 gm/l Indopon T pdr.H/Conc and
2-3 gm/l soda ash

at boil for 20-30 minutes, followed by a hot and cold rinse.

NOTE :

1. All Indothol AS-G combinations are soaped with and addition of 2-3 gm/l Sodium hydrosulphite to develop the true bright shade.
2. Combination of dyeings having low soda boil fastness are to be soaped at 50°C with detergent only.

Fastness :

The highest fastness is represented by figure 5 and the lowest by figure 1, except in the case of light where the highest fastness is taken as 8.

HARIHAR CHEMICALS

(The information contained in this **SHADE CARD** has been provided in good faith and to the best of our knowledge but **WITHOUT WARRANTY**. Consumers are requested to test the products as to their suitability for any application, use or processing)

PATTERNS

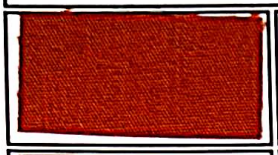
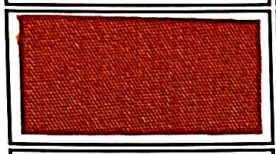
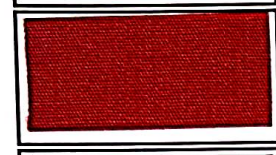
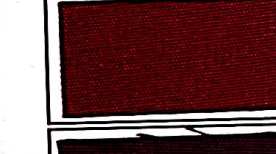
HARIHAR CHEMICALS

NAPHTHOL AS

1%	4%	Product	Kgs. Harithol absorbed per 100 kg. Cotton	Dye bath Conc. in g/l MLR 1:20		Fastness Properties				
				Harithol	Fast Base	Light	Soda boil	Chlorine	Peroxide	Hot pressing
		Fast Yellow GC Base	1.5	3	1.6	5	3	4-5	1-2	2-3
		Fast Orange GC Base	1.5	9	1.8	5	3	4-5	1-2	3-4
		Fast Orange GR Base	1.5	3	1.3	6	3-4	4-5	2	4
		Fast Scarlet G Base	1.5	3	1.4	6	4	5	2-3	3-4
		Fast Scarlet R Base	1.5	3	1.7	5	4	4	2	3
		Fast Scarlet RC Base	1.5	3	2.3	5	4	4	2	3
		Fast Red B Base	1.5	3	1.6	6-7	3-4	4-5	3	3
		Fast Red RC Base	1.5	3	2.0	6	4	4-5	2-3	4
		Fast Red TR Base	1.5	3	2.0	4-5	4	4-5	1	2-3
		Fast Red R Flakes	1.75	2.0	1.5	5	4	4-5	3	2
		Fast Bordeaux GP Base	1.5	3	1.8	6	3	4	3-4	2
		Fast Garnet GBC Base	1.5	3	2.7	6-7	4	4-5	2	4
		Fast Blue B Base	1.0	1.5	1.0	4	4-5	4	3	3-4
		Fast Blue BB Base	1.0	1.5	2.5	5	5	3-4	4	2

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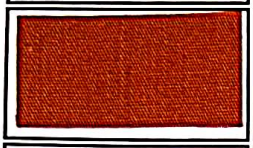
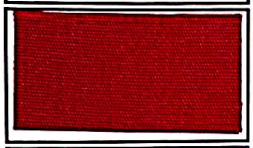
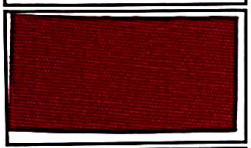
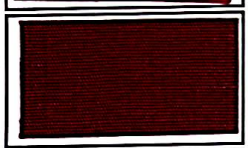
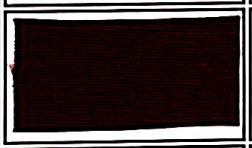
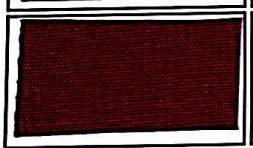
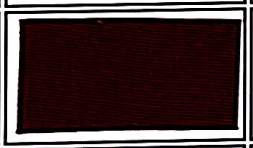
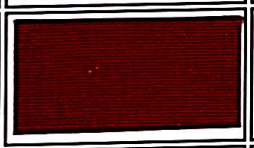
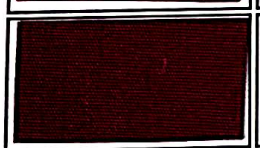
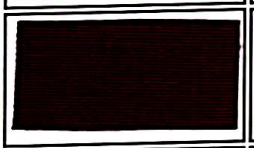
NAPHTHOL AS-BO

1%	4%	Product	Kgs. Harithol absorbed per 100 kg. Cotton	Dye bath Conc. in g/l MLR 1:20		Fastness Properties				
				Harithol	Fast Base	Light	Soda boil	Chlorine	Peroxide	Hot pressing
		Fast Yellow GC Base	1.5	3	1.6	5	3	4-5	1-2	2-3
		Fast Orange GC Base	1.5	9	1.8	5	3	4-5	1-2	3-4
		Fast Orange GR Base	1.5	3	1.3	6	3-4	4-5	2	4
		Fast Scarlet G Base	1.5	3	1.4	6	4	5	2-3	3-4
		Fast Scarlet R Base	1.5	3	1.7	5	4	4	2	3
		Fast Scarlet RC Base	1.5	3	2.3	5	4	4	2	3
		Fast Red B Base	1.5	3	1.6	6-7	3-4	4-5	3	3
		Fast Red RC Base	1.5	3	2.0	6	4	4-5	2-3	4
		Fast Red TR Base	1.5	3	2.0	4-5	4	4-5	1	2-3
		Fast Red R Flakes	1.75	2.0	1.5	5	4	4-5	3	2
		Fast Bordeaux GP Base	1.5	3	1.8	6	3	4	3-4	2
		Fast Garnet GBC Base	1.5	3	2.7	6-7	4	4-5	2	4
		Fast Blue B Base	1.0	1.5	1.0	4	4-5	4	3	3-4
		Fast Blue BB Base	1.0	1.5	2.5	5	5	3-4	4	2

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NAPHTHOL AS-BS

1%	4%	Product	Kgs. Harithol absorbed per 100 kg. Cotton	Dye bath Conc. in g/l MLR 1:20		Fastness Properties				
				Harithol	Fast Base	Light	Soda boil	Chlorine	Peroxide	Hot pressing
		**Fast Yellow GC Base**	1.5	3	1.6	5	3	4-5	1-2	2-3
		**Fast Orange GC Base**	1.5	9	1.8	5	3	4-5	1-2	3-4
		**Fast Orange GR Base**	1.5	3	1.3	6	3-4	4-5	2	4
		**Fast Scarlet G Base**	1.5	3	1.4	6	4	5	2-3	3-4
		**Fast Scarlet R Base**	1.5	3	1.7	5	4	4	2	3
		**Fast Scarlet RC Base**	1.5	3	2.3	5	4	4	2	3
		**Fast Red B Base**	1.5	3	1.6	6-7	3-4	4-5	3	3
		**Fast Red RC Base**	1.5	3	2.0	6	4	4-5	2-3	4
		**Fast Red TR Base**	1.5	3	2.0	4-5	4	4-5	1	2-3
		**Fast Red R Flakes**	1.75	2.0	1.5	5	4	4-5	3	2
		**Fast Bordeaux GP Base**	1.5	3	1.8	6	3	4	3-4	2
		**Fast Garnet GBC Base**	1.5	3	2.7	6-7	4	4-5	2	4
		**Fast Blue B Base**	1.0	1.5	1.0	4	4-5	4	3	3-4
		**Fast Blue BB Base**	1.0	1.5	2.5	5	5	3-4	4	2

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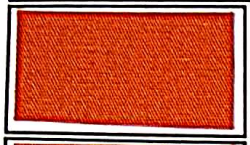
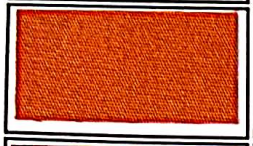
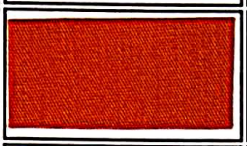
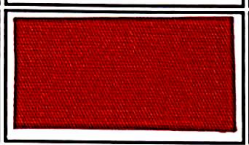
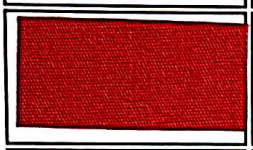
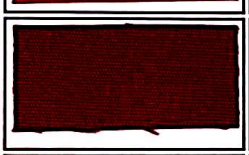
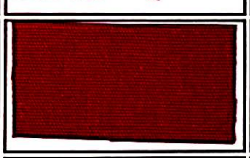
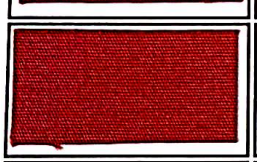
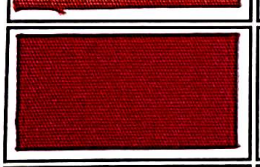
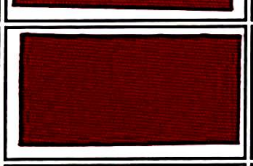
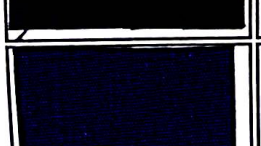
NAPHTHOL AS-G

1%	4%	Product	Kgs. Harithol absorbed per 100 kg. Cotton	Dye bath Conc. in g/l MLR 1:20		Fastness Properties				
				Harithol	Fast Base	Light	Soda boil	Chlorine	Peroxide	Hot pressing
		Fast Yellow GC Base	1.5	3	1.6	5	3	4-5	1-2	2-3
		Fast Orange GC Base	1.5	9	1.8	5	3	4-5	1-2	3-4
		Fast Orange GR Base	1.5	3	1.3	6	3-4	4-5	2	4
		Fast Scarlet G Base	1.5	3	1.4	6	4	5	2-3	3-4
		Fast Scarlet R Base	1.5	3	1.7	5	4	4	2	3
		Fast Scarlet RC Base	1.5	3	2.3	5	4	4	2	3
		Fast Red B Base	1.5	3	1.6	6-7	3-4	4-5	3	3
		Fast Red RC Base	1.5	3	2.0	6	4	4-5	2-3	4
		Fast Red TR Base	1.5	3	2.0	4-5	4	4-5	1	2-3
		Fast Red R Flakes	1.75	2.0	1.5	5	4	4-5	3	2
		Fast Bordeaux GP Base	1.5	3	1.8	6	3	4	3-4	2
		Fast Garnet GBC Base	1.5	3	2.7	6-7	4	4-5	2	4
		Fast Blue B Base	1.0	1.5	1.0	4	4-5	4	3	3-4
		Fast Blue BB Base	1.0	1.5	2.5	5	5	3-4	4	2

HARIHAR CHEMICALS

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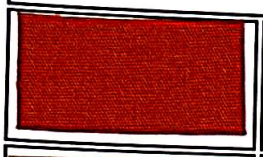
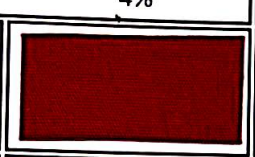
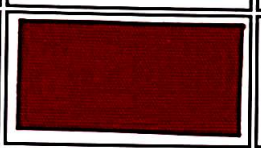
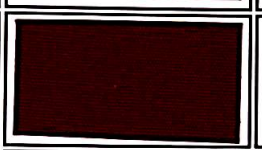
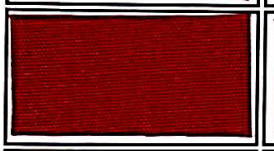
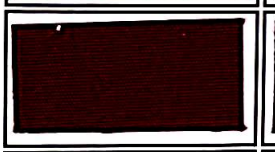
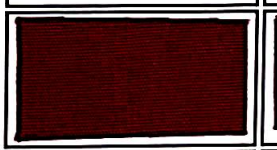
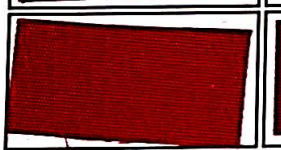
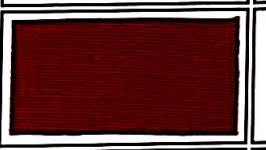
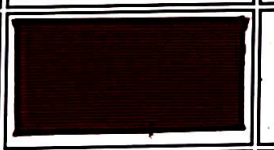
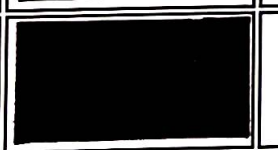
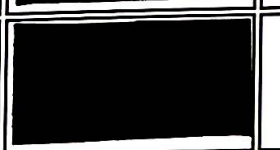
NAPHTHOL AS-OL

1%	4%	Product	Kgs. Harithol absorbed per 100 kg. Cotton	Dye bath Conc. in g/l MLR 1:20		Fastness Properties				
				Harithol	Fast Base	Light	Soda boil	Chlorine	Peroxide	Hot pressing
		**Fast Yellow GC Base**	1.5	3	1.6	5	3	4-5	1-2	2-3
		**Fast Orange GC Base**	1.5	9	1.8	5	3	4-5	1-2	3-4
		**Fast Orange GR Base**	1.5	3	1.3	6	3-4	4-5	2	4
		**Fast Scarlet G Base**	1.5	3	1.4	6	4	5	2-3	3-4
		**Fast Scarlet R Base**	1.5	3	1.7	5	4	4	2	3
		**Fast Scarlet RC Base**	1.5	3	2.3	5	4	4	2	3
		**Fast Red B Base**	1.5	3	1.6	6-7	3-4	4-5	3	3
		**Fast Red RC Base**	1.5	3	2.0	6	4	4-5	2-3	4
		**Fast Red TR Base**	1.5	3	2.0	4-5	4	4-5	1	2-3
		**Fast Red R Flakes**	1.75	2.0	1.5	5	4	4-5	3	2
		**Fast Bordeaux GP Base**	1.5	3	1.8	6	3	4	3-4	2
		**Fast Garnet GBC Base**	1.5	3	2.7	6-7	4	4-5	2	4
		**Fast Blue B Base**	1.0	1.5	1.0	4	4-5	4	3	3-4
		**Fast Blue BB Base**	1.0	1.5	2.5	5	5	3-4	4	2

HARIHAR CHEMICALS

HARIHAR CHEMICALS

NAPHTHOL AS-SW

1%	4%	Product	Kgs. Harithol absorbed per 100 kg. Cotton	Dye bath Conc. in g/l MLR 1:20		Fastness Properties				
				Harithol	Fast Base	Light	Soda boil	Chlorine	Peroxide	Hot pressing
		Fast Yellow GC Base	1.5	3	1.6	5	3	4-5	1-2	2-3
		Fast Orange GC Base	1.5	9	1.8	5	3	4-5	1-2	3-4
		Fast Orange GR Base	1.5	3	1.3	6	3-4	4-5	2	4
		Fast Scarlet G Base	1.5	3	1.4	6	4	5	2-3	3-4
		Fast Scarlet R Base	1.5	3	1.7	5	4	4	2	3
		Fast Scarlet RC Base	1.5	3	2.3	5	4	4	2	3
		Fast Red B Base	1.5	3	1.6	6-7	3-4	4-5	3	3
		Fast Red RC Base	1.5	3	2.0	6	4	4-5	2-3	4
		Fast Red TR Base	1.5	3	2.0	4-5	4	4-5	1	2-3
		Fast Red R Flakes	1.75	2.0	1.5	5	4	4-5	3	2
		Fast Bordeaux GP Base	1.5	3	1.8	6	3	4	3-4	2
		Fast Garnet GBC Base	1.5	3	2.7	6-7	4	4-5	2	4
		Fast Blue B Base	1.0	1.5	1.0	4	4-5	4	3	3-4
		Fast Blue BB Base	1.0	1.5	2.5	5	5	3-4	4	2

HARIHAR CHEMICALS

NAPHTHOL AS-D

1%	4%	Product	Kgs. Harithol absorbed per 100 kg. Cotton	Dye bath Conc. in g/l MLR 1:20		Fastness Properties				
				Harithol	Fast Base	Light	Soda boil	Chlorine	Peroxide	Hot pressing
		Fast Yellow GC Base	1.5	3	1.6	5	3	4-5	1-2	2-3
		Fast Orange GC Base	1.5	9	1.8	5	3	4-5	1-2	3-4
		Fast Orange GR Base	1.5	3	1.3	6	3-4	4-5	2	4
		Fast Scarlet G Base	1.5	3	1.4	6	4	5	2-3	3-4
		Fast Scarlet R Base	1.5	3	1.7	5	4	4	2	3
		Fast Scarlet RC Base	1.5	3	2.3	5	4	4	2	3
		Fast Red B Base	1.5	3	1.6	6-7	3-4	4-5	3	3
		Fast Red RC Base	1.5	3	2.0	6	4	4-5	2-3	4
		Fast Red TR Base	1.5	3	2.0	4-5	4	4-5	1	2-3
		Fast Red R Flakes	1.75	2.0	1.5	5	4	4-5	3	2
		Fast Bordeaux GP Base	1.5	3	1.8	6	3	4	3-4	2
		Fast Garnet GBC Base	1.5	3	2.7	6-7	4	4-5	2	4
		Fast Blue B Base	1.0	1.5	1.0	4	4-5	4	3	3-4
		Fast Blue BB Base	1.0	1.5	2.5	5	5	3-4	4	2

HARIHAR CHEMICALS

HARIHAR CHEMICALS

NAPHTHOL AS-TR

1%	4%	Product	Kgs. Harithol absorbed per 100 kg. Cotton	Dye bath Conc. in g/l MLR 1:20		Fastness Properties				
				Harithol	Fast Base	Light	Soda boil	Chlorine	Peroxide	Hot pressing
		Fast Yellow GC Base	1.5	3	1.6	5	3	4-5	1-2	2-3
		Fast Orange GC Base	1.5	9	1.8	5	3	4-5	1-2	3-4
		Fast Orange GR Base	1.5	3	1.3	6	3-4	4-5	2	4
		Fast Scarlet G Base	1.5	3	1.4	6	4	5	2-3	3-4
		Fast Scarlet R Base	1.5	3	1.7	5	4	4	2	3
		Fast Scarlet RC Base	1.5	3	2.3	5	4	4	2	3
		Fast Red B Base	1.5	3	1.6	6-7	3-4	4-5	3	3
		Fast Red RC Base	1.5	3	2.0	6	4	4-5	2-3	4
		Fast Red TR Base	1.5	3	2.0	4-5	4	4-5	1	2-3
		Fast Red R Flakes	1.75	2.0	1.5	5	4	4-5	3	2
		Fast Bordeaux GP Base	1.5	3	1.8	6	3	4	3-4	2
		Fast Garnet GBC Base	1.5	3	2.7	6-7	4	4-5	2	4
		Fast Blue B Base	1.0	1.5	1.0	4	4-5	4	3	3-4
		Fast Blue BB Base	1.0	1.5	2.5	5	5	3-4	4	2

HARIHAR CHEMICALS

NAPHTHOL AS-E

1%	4%	Product	Kgs. Harithol absorbed per 100 kg. Cotton	Dye bath Conc. in g/l MLR 1:20		Fastness Properties				
				Harithol	Fast Base	Light	Soda boil	Chlorine	Peroxide	Hot pressing
		Fast Yellow GC Base	1.5	3	1.6	5	3	4-5	1-2	2-3
		Fast Orange GC Base	1.5	9	1.8	5	3	4-5	1-2	3-4
		Fast Orange GR Base	1.5	3	1.3	6	3-4	4-5	2	4
		Fast Scarlet G Base	1.5	3	1.4	6	4	5	2-3	3-4
		Fast Scarlet R Base	1.5	3	1.7	5	4	4	2	3
		Fast Scarlet RC Base	1.5	3	2.3	5	4	4	2	3
		Fast Red B Base	1.5	3	1.6	6-7	3-4	4-5	3	3
		Fast Red RC Base	1.5	3	2.0	6	4	4-5	2-3	4
		Fast Red TR Base	1.5	3	2.0	4-5	4	4-5	1	2-3
		Fast Red R Flakes	1.75	2.0	1.5	5	4	4-5	3	2
		Fast Bordeaux GP Base	1.5	3	1.8	6	3	4	3-4	2
		Fast Garnet GBC Base	1.5	3	2.7	6-7	4	4-5	2	4
		Fast Blue B Base	1.0	1.5	1.0	4	4-5	4	3	3-4
		Fast Blue BB Base	1.0	1.5	2.5	5	5	3-4	4	2

HARIHAR CHEMICALS